Wave Packets in Motion: Phase and Group Velocity

PHYS 2203 — Introductory Physics III

November 2, 2026

Overview

Last lecture used wave packets, built from plane waves in Lecture 26, to arrive at the uncertainty principle. After a short recap of matter waves and wave packets, this lecture asks how such a packet moves. Each plane wave travels at its own speed, the phase velocity, while the packet as a whole, where the particle is likely to be found, travels at the group velocity. We find the group velocity from a simple group of two waves, then show how a wave packet evolves under the Schrödinger equation. The exact result for a Gaussian packet shows that it moves at the particle's velocity and spreads out as it moves, quickly for an electron and imperceptibly for a baseball.

1 Matter Waves and Wave Packets: A Recap

A matter wave with wave number k and angular frequency ω describes a particle with momentum and energy

$$p = \hbar k, \qquad E = \hbar\omega. \tag{1}$$

The plane wave $\Psi = Ae^{i(kx-\omega t)}$ solves the free-particle Schrödinger equation provided

$$\omega = \frac{\hbar k^2}{2m}, \qquad \text{that is,} \qquad E = \frac{p^2}{2m}. \tag{2}$$

Its crests move at $f\lambda = \omega/k = E/p$, which is not the particle's speed p/m. A plane wave has a definite momentum, but the particle is equally likely to be anywhere.

A localized particle is described instead by a wave packet, a sum of plane waves with amplitudes $A(k)$, $\psi(x) = \int A(k)\, e^{ikx}\, \mathrm{d}k$. For the Gaussian packet

$$\psi(x) = C\, e^{-(x/2\varepsilon)^2}\, e^{ik_0 x}, \tag{3}$$

the Fourier transform is another Gaussian, centered on k_0 (Lecture 26),

$$A(k) = \frac{C\varepsilon}{\sqrt{\pi}}\, e^{-\varepsilon^2(k-k_0)^2}, \tag{4}$$

so the packet describes a particle with momentum near $p_0 = \hbar k_0$. Its widths are $\Delta x = \varepsilon$ and $\Delta k = 1/2\varepsilon$, so a narrow packet contains a wide range of wave numbers. This lecture asks how such a packet moves.

2 Phase and Group Velocities

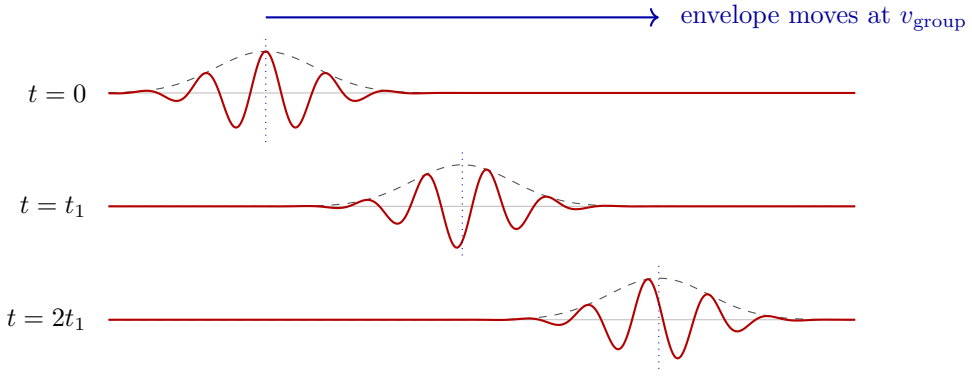


Figure 1: A traveling wave group at three times. The envelope (dashed) moves at the group velocity, and the dotted lines mark its center. The individual crests move at a different speed, the phase velocity, and slip through the envelope.

- **Phase velocity:** the speed of an individual crest of one plane wave,

$$\boxed{v_{\text{phase}} = f\lambda = \frac{\omega}{k}} \tag{5}$$

- **Group velocity:** the speed of the envelope, where $|\Psi|^2$ is large. This is where the particle is likely to be, so it is the speed that matters physically.

3 A Simple Wave Group

Add two plane waves whose wave numbers and frequencies differ slightly from central values k_0 and ω_0:

$$k_{1,2} = k_0 \pm \mathrm{d}k, \qquad \omega_{1,2} = \omega_0 \pm \mathrm{d}\omega. \tag{6}$$

$$\Psi(x,t) = Ae^{i(k_1 x - \omega_1 t)} + Ae^{i(k_2 x - \omega_2 t)} = Ae^{i(k_0 x - \omega_0 t)}\left(e^{+i(\mathrm{d}k\, x - \mathrm{d}\omega\, t)} + e^{-i(\mathrm{d}k\, x - \mathrm{d}\omega\, t)}\right), \tag{7}$$

$$\boxed{\Psi(x,t) = \underbrace{Ae^{i(k_0 x - \omega_0 t)}}_{\text{moves at } \omega_0/k_0}\ \underbrace{2\cos(\mathrm{d}k\, x - \mathrm{d}\omega\, t)}_{\text{envelope, moves at } \mathrm{d}\omega/\mathrm{d}k}} \tag{8}$$

This is the beat pattern of Lecture 26, now moving. The fast oscillation moves at the phase velocity ω_0/k_0. The envelope moves at $\mathrm{d}\omega/\mathrm{d}k$. The probability density keeps only the envelope:

$$|\Psi(x,t)|^2 = 4A^2\cos^2(\mathrm{d}k\, x - \mathrm{d}\omega\, t), \tag{9}$$

so the probability moves at

$$\boxed{v_{\text{group}} = \left.\frac{\mathrm{d}\omega}{\mathrm{d}k}\right|_{k_0}} \tag{10}$$

The group velocity depends not on ω and k themselves but on how ω changes with k. The same is true for a packet built from a continuous range of k, as we will see in Section 5.

4 Wave Packets and the Schrödinger Equation

In Lecture 26 the Gaussian packet was written at a single instant, $\psi(x) = \int A(k)\, e^{ikx}\, \mathrm{d}k$. It is a starting wave function, not yet a solution of the Schrödinger equation, which involves time. To let it evolve, give each plane wave its own time dependence. From Lecture 25, $e^{i(kx-\omega t)}$ solves the free-particle Schrödinger equation when $\omega = \hbar k^2/2m$. The equation is linear, so a sum of solutions is again a solution:

$$\boxed{\Psi(x,t) = \int_{-\infty}^{+\infty} A(k)\, e^{i(kx-\omega(k)t)}\, \mathrm{d}k, \qquad \omega(k) = \frac{\hbar k^2}{2m}} \tag{11}$$

To check that this is a solution, take the derivatives. The integral is over k, so the derivatives with respect to x and t act only on the exponential inside it:

$$\frac{\partial^2 \Psi}{\partial x^2} = \int A(k)\, (ik)^2\, e^{i(kx-\omega t)}\, \mathrm{d}k = -\int k^2 A(k)\, e^{i(kx-\omega t)}\, \mathrm{d}k, \tag{12}$$

$$\frac{\partial \Psi}{\partial t} = \int A(k)\, (-i\omega)\, e^{i(kx-\omega t)}\, \mathrm{d}k. \tag{13}$$

Substituting into the two sides of the Schrödinger equation,

$$-\frac{\hbar^2}{2m}\frac{\partial^2 \Psi}{\partial x^2} = \int \frac{\hbar^2 k^2}{2m}\, A(k)\, e^{i(kx-\omega t)}\, \mathrm{d}k, \qquad i\hbar\,\frac{\partial \Psi}{\partial t} = \int \hbar\omega\, A(k)\, e^{i(kx-\omega t)}\, \mathrm{d}k. \tag{14}$$

The two integrals are equal term by term, because $\hbar^2 k^2/2m = \hbar\omega$ for every k. So $\Psi(x,t)$ is a solution. At $t = 0$ every factor $e^{-i\omega t}$ equals 1, and Ψ reduces to the starting packet $\psi(x)$.

For the Gaussian packet this integral can be evaluated exactly (Section 5). The result shows that the envelope moves at the group velocity, just as for the simple wave group, and that it slowly spreads.

5 Dispersion: Wave Packets Spread

For matter waves $\omega = \hbar k^2/2m$, so plane waves with different wave numbers travel at different phase velocities, $\omega/k = \hbar k/2m$. As time goes on, the plane waves making up a packet get out of step, and the packet spreads out. This is **dispersion**, and for matter waves it happens *even in vacuum.* For a Gaussian packet of initial width ε, inserting $A(k)$ from Eq. (4) and $\omega = \hbar k^2/2m$ into $\Psi(x,t) = \int A(k)\, e^{i(kx-\omega t)}\, \mathrm{d}k$ gives a Gaussian integral that can be done exactly. Writing $v_0 = \hbar k_0/m = p_0/m$ for the velocity of a classical particle with momentum p_0, the squared magnitude of the result is

$$|\Psi(x,t)|^2 = \frac{C^2}{\sqrt{1 + \hbar^2 t^2/4m^2\varepsilon^4}}\, \exp\left[\frac{-(x - v_0 t)^2}{2\varepsilon^2\,(1 + \hbar^2 t^2/4m^2\varepsilon^4)}\right]. \tag{15}$$

- The peak slides along at v_0, the particle's velocity. This is the group velocity: for $\omega = \hbar k^2/2m$, $\mathrm{d}\omega/\mathrm{d}k = \hbar k_0/m = v_0$ at k_0, just as the simple wave group predicts.
- The width grows from ε to $\varepsilon\sqrt{1 + \hbar^2 t^2/4m^2\varepsilon^4}$, and the height falls, so the total probability stays 1.
- The spreading rate is proportional to $1/\varepsilon^2$: the narrower the packet starts, the faster it spreads. A narrow packet contains a wide range of wave numbers (Lecture 27), and so a wide range of velocities.

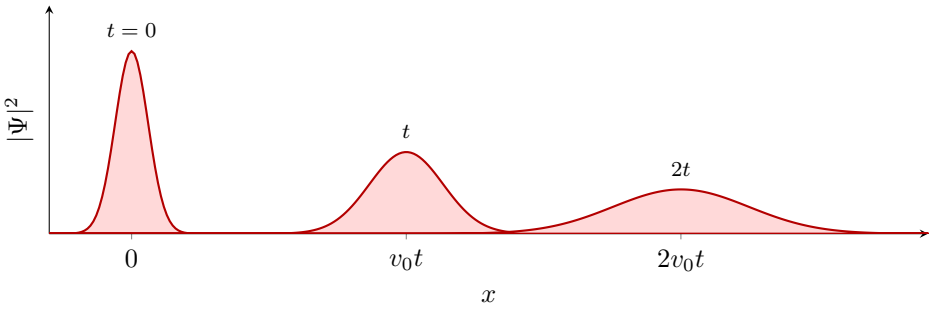


Figure 2: The probability density of a free particle at three equally spaced times. The packet moves at the group velocity and spreads, so the particle's position becomes more and more uncertain.

Example: an electron and a baseball. The width doubles when $\hbar t/2m\varepsilon^2 = \sqrt{3}$, after a time $t = 2\sqrt{3}\, m\varepsilon^2/\hbar$. For an electron localized to $\varepsilon = 0.1$ nm, about the size of an atom,

$$t = \frac{2\sqrt{3}\,(9.11 \times 10^{-31}\ \text{kg})(1.0 \times 10^{-10}\ \text{m})^2}{1.055 \times 10^{-34}\ \text{J}\cdot\text{s}} = 3.0 \times 10^{-16}\ \text{s}. \tag{16}$$

A free electron localized this well spreads almost immediately. For a 0.145 kg baseball located to within $\varepsilon = 1$ mm,

$$t = \frac{2\sqrt{3}\,(0.145\ \text{kg})(1.0 \times 10^{-3}\ \text{m})^2}{1.055 \times 10^{-34}\ \text{J}\cdot\text{s}} = 4.8 \times 10^{27}\ \text{s}, \tag{17}$$

about 10^{10} times the age of the universe. The baseball's packet never spreads noticeably, and it behaves as a classical particle.